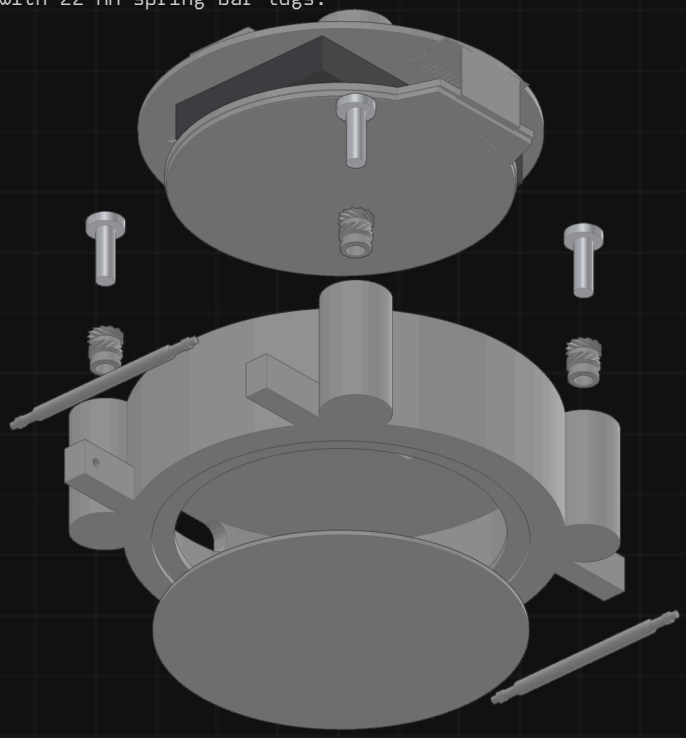
Rev A | 2026-10-24 | UHD-WATCH-AG

ASSEMBLY GUIDE

Wearable Watch

Reference smartwatch: Raytac MDBT50Q-1MV2 (nRF52840, BLE), 1.28-inch round 240 x 240 display (ER-TFT1.28-1, GC9A01), BMI270 IMU, MAX30102 heart rate, DRV2605L with a VG0832012 LRA, MAX17048 fuel gauge, MCP73831 charger from USB-C, 200 mAh LiPo (LP402030), two side buttons, printed PETG case with 22 mm spring-bar lugs.



Exploded along each joint's axis in fastener-stack order (7 mates, 8 hardware parts, battery by placement hint).

STEPS	MATES	HARDWARE	PARTS
7	7	8	71
mechanical and wiring	from assemble()	screws, spacers, standoffs, nuts	45 BOM lines

HOW THIS GUIDE IS MADE

Steps, parts and hardware come from the UHD model: mates and fastener stacks from `assemble()`, wiring from `wiringChecklist`. The coverage table at the end lists where every mate, fastener and wire appears.

[A] Assumption: the model marks this value as an assumption (assumption trait)

[D] Derived: computed from model values; formula in the provenance appendix

Unmarked values are cited from the part's sources (sources.json) or are design values of the model.

01 Before you begin

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TOOLS

- Soldering iron, solder, heat-shrink

Derived from the fasteners in the model: hex keys from each screw's stated socket or the ISO 4762 socket size for its thread, drivers and spanners from each nut's and standoff's across-flats.

READING THE FIGURES

Already assembled
Fitted in this step
Hardware for this step
------ Assembly axis

Parts are drawn exploded along each joint's axis in stacking order, from the model's fastener stacks and mate gaps.

NOTE Each step's torque line gives the tightening torque of the part turned, from the harness's fastener stack. 1 of them are **assumptions** (marked A): no frame, motor or DJI source states a torque, so they are derated from supplier tables for the screw grade into steel. Threadlocker is only advised where a harness says so (steel into aluminium).

02 Kit contents

DOCS/WEARABLE-WATCH-BOM.JSON



×1

Watch main board (assembled)

watch-main-board

no CAD

×1

Cell foam pad, 0.25 mm double-sided

watch-cell-foam-pad

no CAD

×1

Clear UV-curing adhesive for the window covers

watch-cover-adhesive

no CAD

×1

Display adhesive ring, 0.25 mm

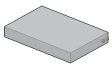
watch-display-adhesive-ring



×2

Esslinger 82.2225 spring bar, 22 mm, ø1.50 mm, double flange

esslinger-82-2225-spring-bar



×1

LiPol LP402030 200 mAh 3.7 V LiPo with PCM

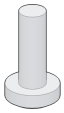
lipol-lp402030

no CAD

×1

LRA adhesive pad, 0.25 mm, ø8 mm

watch-lra-adhesive-pad



×4

M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless

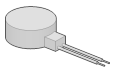
iso-7045-m2x6-a2



×4

ruthex RX-M2x4 brass heat-set insert, M2 × 4.0 mm

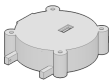
ruthex-rx-m2x4



×1

Vybronics VG0832012 8 mm Z-axis coin LRA

vybronics-vg0832012



×1

Watch case

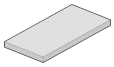
watch-case



×1

Display lens, 38 × 0.5 mm PMMA

watch-display-lens



×1

Heart-rate window cover, 4.0 × 8.2 × 0.5 mm PMMA

watch-hr-cover



×1

Cell leads

watch-battery-leads



×1

LRA leads

watch-lra-leads

Quantities from the generated BOM; a circuit board is one item, assembled by the fab (its parts are in the BOM, not at the bench); drawings are the parts' CAD artifacts, not to scale.

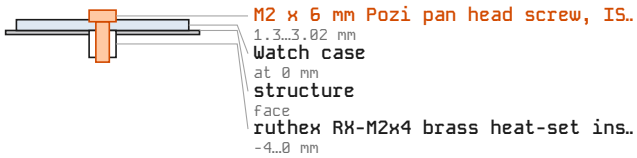
03 Hardware

FASTENER STACKS

Every screw, spacer, standoff and nut is placed by a harness's `fastenerStack`. The sections below are drawn from those positions and the mate gaps: one hole of each joint, to one scale, millimetres from the structure face.

HARNESS	PARTS PLACED	TORQUE	USED IN
Back cover screws joint back_cover	4 × ruthex RX-M2x4 brass heat-set insert, M2 × 4.0 mm 4 × M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless	0.1 N·m[A] M2 × 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless	Step 6

Back cover screws watch-back-cover-screws



1 Fit the lens, the heart-rate cover and the plungers

mates: lens_seat, hr_cover_seat



Ⓐ ×1
Display lens, 38 × 0.5 mm PMMA



Ⓑ ×1
Heart-rate window cover, 4.0 × 8.2 × 0.5 mm PMMA

Ⓒ ×1
Clear UV-curing adhesive for the window covers

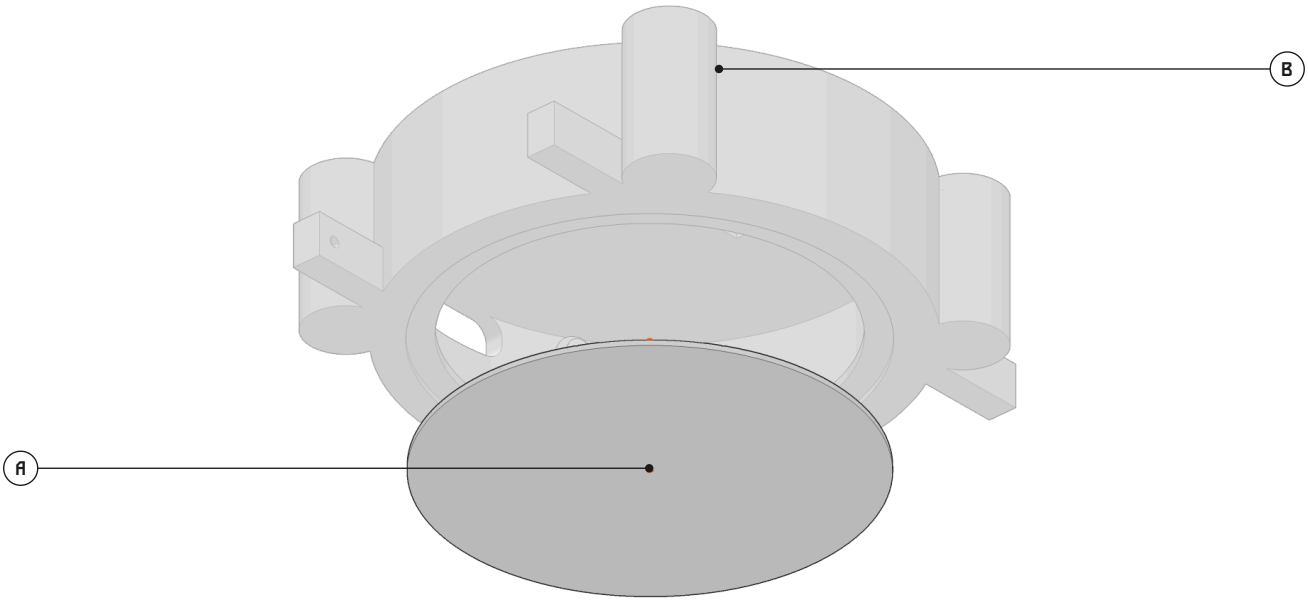


Figure 1 Fit the lens, the heart-rate cover and the plungers: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Seat the Display lens, 38 × 0.5 mm PMMA onto the display window and lens.
2. Seat the Heart-rate window cover, 4.0 × 8.2 × 0.5 mm PMMA onto the heart-rate window.
3. Fit the Clear UV-curing adhesive for the window covers.
4. Push the two printed plungers (plunger_button_1, plunger_button_2) into their bores in the case wall from inside, flange inwards; each must slide freely.

Step 1 of 7

TIP Peel the protective film from the inner face only. Put a thin bead of adhesive on each pocket's ledge, press the cover in flush and cure it before anything goes inside the case.

CAUTION Keep adhesive off the apertures: on the heart-rate window it would scatter the sensor's light (OQ-W7).

2 Solder the cell leads to J3

mates: board/j3_site · links: cell_leads, cell_pads



Figure 2 Pad-to-pad connections from the stored links (wiringChecklist); crossovers in accent.

1. Seat the Cell lead solder pads (2) onto the on the board copper.
2. Lay the cell flat beside the board's +y edge, PCM end towards J3 and no more than 3 mm past the board's edge, so its leads reach J3 on the board's rear face.
3. Connect Cell leads, Cell leads on J3 as in the table (4 connections).

LINK	FROM	TO	TERMINATION	DRC
Cell leads	cell **+ · -**	Cell leads **P1 · P2**	integral	CONFIGURED
Cell leads on J3	Cell leads **P1 · P2**	board **1 · 2**	solder_leads	CONFIGURED

Step 2 of 7

⚠ **CAUTION** Red (+) to pad 1, black (-) to pad 2. Cut and strip one lead at a time so the two never touch: the cell is charged and its PCM is the only protection.

TIP Cut the leads to the length in the BOM (installed length plus 1.5 mm stripped) before stripping: the cell then folds under the board with no slack to spare.

NOTE The cell's own leads: cut to 9 mm, strip 1.5 mm, tin, and solder + (red) to J3 pad 1 and - (black) to pad 2 with the cell lying beside the board's +y edge, then fold the cell under the board's rear face. Never let the bare ends touch: the cell's PCM trips at 1-2.5 A, but a short still heats the leads. — watch-battery-leads

3 Fold the cell and the display, and seat the board in the case



Ⓐ ×1
Watch main board



Ⓑ ×1
LiPol LP402030 200 mAh 3.7 V LiPo with PCM

Ⓒ ×1
Cell foam pad, 0.25 mm double-sided

Ⓓ ×1
Display adhesive ring, 0.25 mm

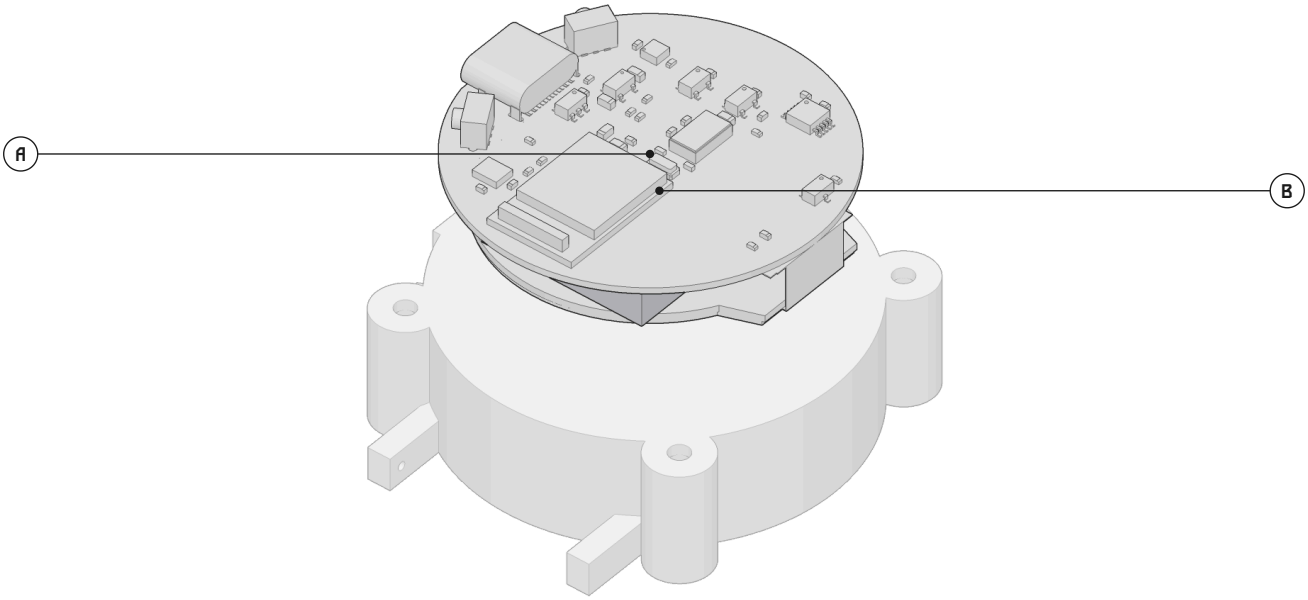


Figure 3 Fold the cell and the display, and seat the board in the case: shown lifted above its seat.

1. Place the Watch main board as shown. It has no mechanical interface in the model; the figure places it by .protoboard/display.json: mid-plane at z 6.68: the display it carries on its rear face (soldered by the FPC, 4.78 mm below the rear copper) puts the viewing face at z 0 on the lens; its components face the back cover; the USB-C mouth and the switch stems line up with the case wall at this height (the case is generated around it).
2. Place the LiPol LP402030 200 mAh 3.7 V LiPo with PCM as shown. It has no mechanical interface in the model; the figure places it by .protoboard/display.json: between the display and the board, PCM end towards 12 o'clock as seen from the back (+y), on 0.25 mm foam tape on the display's rear face, 0.53 mm clear of the board for the cell to swell (D-W15); held by the foam tape (D-W6).
3. Fit the Cell foam pad, 0.25 mm double-sided.
4. Fit the Display adhesive ring, 0.25 mm.
5. Fold the cell under the board's rear face on its leads, PCM end towards 12 o'clock, and stick it to the display's rear face with the foam pad.
6. Fold the display over the cell on its FPC, viewing face out.
7. Stick the adhesive ring to the display's front rim, outside its visual area, and lower the board and display into the case: display face onto the floor, USB-C into its opening, button stems behind their plungers.

Step 3 of 7

⚠ **CAUTION** Do not crease the display's FPC: it bends once, past the cell's edge, as it left the board unit.

4 Solder the LRA leads to J4

mates: board/j4_site · links: lra_wires, lra_pads



Figure 4 Pad-to-pad connections from the stored links (wiringChecklist); crossovers in accent.

1. Seat the LRA lead solder pads (2) onto the on the board copper.

2. Connect LRA leads, LRA leads on J4 as in the table (4 connections).

LINK	FROM	TO	TERMINATION	DRC
LRA leads	lra **+ · -**	LRA leads **P1 · P2**	integral	CONFIGURED
LRA leads on J4	LRA leads **P1 · P2**	board **1 · 2**	solder_leads	CONFIGURED

Step 4 of 7

TIP Red (+) to pad 1, blue (-) to pad 2, with the LRA lying in the board's free region (x 5.7-14.5 mm). An LRA is not polarised in use, but keeping the order makes the test record comparable.

NOTE The LRA's own leads: solder + (red) to J4 pad 1 and - (blue) to pad 2. The LRA is bonded to the back cover with its can face (the Z-axis vibration couples into the wrist). — watch-lra-leads

5 Put the LRA in place for the back cover



Ⓐ ×1
Vybronics VG0832012 8 mm Z-axis coin LRA

Ⓑ ×1
LRA adhesive pad, 0.25 mm, ø8 mm

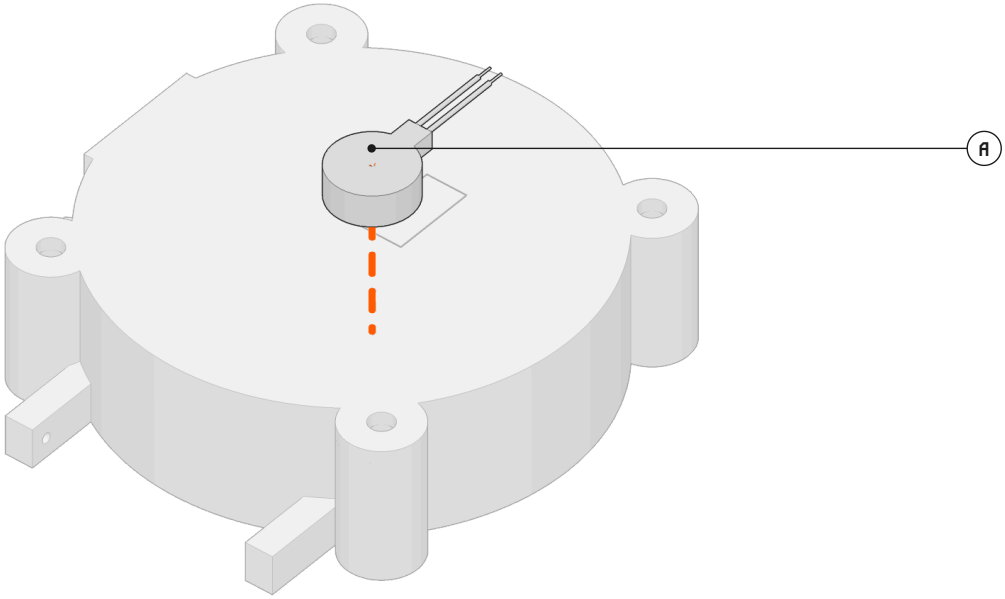


Figure 5 Put the LRA in place for the back cover: shown lifted above its seat.

1. Place the Vybronics VG0832012 8 mm Z-axis coin LRA as shown. It has no mechanical interface in the model; the figure places it by .protoboard/display.json: can face up against the back cover through a 0.25 mm adhesive pad (z 10.39 to the cover's inner face at 10.64), leads towards J4 (+y), in the region of the board kept free of parts (x 5.7-14.5, y -13.5..7.2).
2. Fit the LRA adhesive pad, 0.25 mm, ø8 mm.

Step 5 of 7

TIP Stick the adhesive pad on the LRA's can face and leave its liner on until the back cover goes on: the cover picks the LRA up as it closes. Leads towards J4 (12 o'clock), clear of the heart-rate window.

6 Press the inserts and close the back cover

mates: back_cover



A ×4

ruthex RX-M2x4 brass heat-set insert, M2 x 4.0 mm



B ×4

M2 x 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless

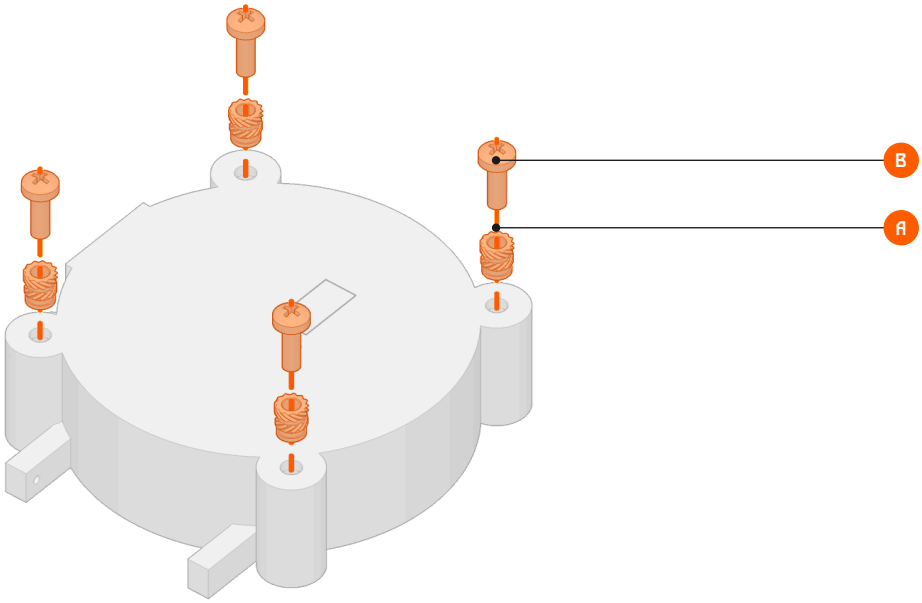


Figure 6 Press the inserts and close the back cover: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Fit 4 × ruthex RX-M2x4 brass heat-set insert, M2 x 4.0 mm on the back cover joint (4 x M2).
2. Seat the Watch case onto the back cover joint (4 x M2).
3. Drive 4 × M2 x 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless from above, head down.
4. Remove the LRA pad's liner, then lower the back cover so the pad meets the cover's inner face, heart-rate window over U3.

Torque 0.1 N·m[A] M2 x 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless **Step** 6 of 7

TIP Press the four inserts flush with the joint face before the first assembly (iron at about 220 °C for PETG). Tighten the screws in a cross pattern.

NOTE Press the inserts into the case columns with a soldering iron and insert tip (about 220 °C for PETG, ruthex's guidance), flush with the joint face; then close the back cover with the screws. The case provider sizes the columns, insert holes and screw reach from these parts (enclosure-cad report, back_joint). — watch-back-cover-screws

7 Fit the spring bars

mates: bar_12_lugs, bar_6_lugs



Ⓐ ×2

Esslinger 82.2225 spring bar, 22 mm, ø1.50 mm, double flange

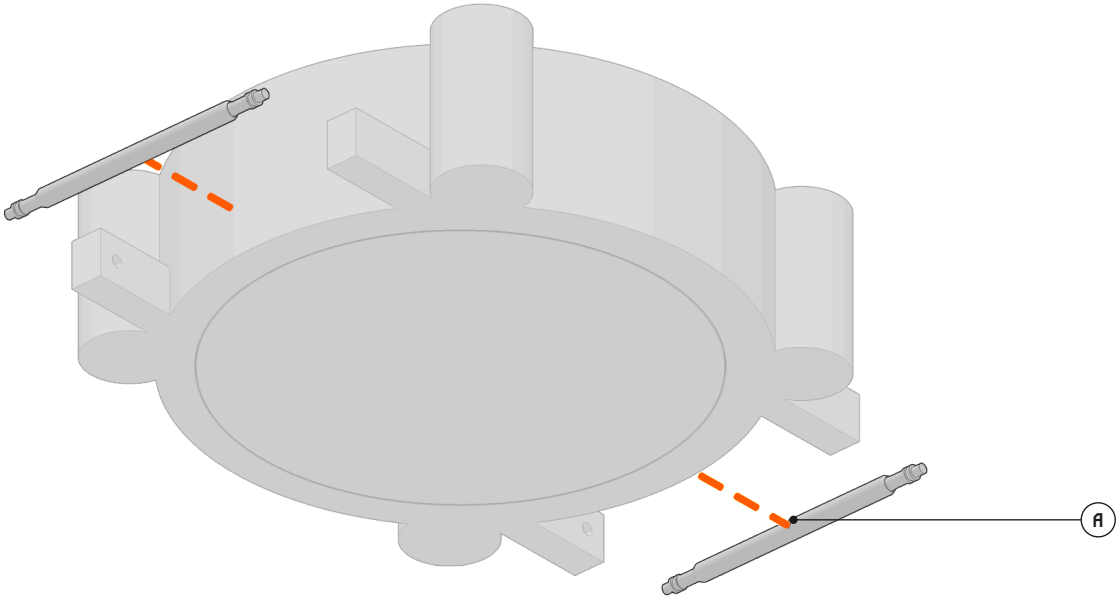


Figure 7 Fit the spring bars: parts exploded along each joint's axis in stacking order; lettered as in the parts strip.

1. Seat a Esslinger 82.2225 spring bar, 22 mm, ø1.50 mm, double flange onto each lugs at 12 o'clock. Repeat on all 2.

Step 7 of 7

11 Coverage

ASSEMBLE() · WIRINGCHECKLIST

Every rigid mate, every hardware placement and every electrical link in the model, and the step that covers it. The verifier fails the guide if any row is not covered.

MATES

MATE	INTERFACES	HARNESS	STEP
lens_seat	case.display_window ↔ lens.seat	—	1 · Fit the lens, the heart-rate cover and the plungers
hr_cover_seat	case.hr_window ↔ hr_cover.seat	—	1 · Fit the lens, the heart-rate cover and the plungers
back_cover	case.back_joint ↔ case.back_joint	back_screws	6 · Press the inserts and close the back cover
bar_12_lugs	case.lugs_12 ↔ bar_12.strap_bar	—	7 · Fit the spring bars
bar_6_lugs	case.lugs_6 ↔ bar_6.strap_bar	—	7 · Fit the spring bars
board/j3_site	board/j3.board_mount ↔ board.site_j3	—	2 · Solder the cell leads to J3
board/j4_site	board/j4.board_mount ↔ board.site_j4	—	4 · Solder the LRA leads to J4

HARDWARE PLACEMENTS

HARNESS / CHILD	PARTS	JOINT	STEP
back_screws/inserts	4 × ruthex RX-M2x4 brass heat-set insert, M2 x 4.0 mm	back_cover	6 · Press the inserts and close the back cover
back_screws/screws	4 × M2 x 6 mm Pozi pan head screw, ISO 7045 / DIN 7985 Z, A2 stainless	back_cover	6 · Press the inserts and close the back cover

WIRING LINKS

LINK	NAME	HARNESS	STEP
cell_leads	Cell leads	—	2 · Solder the cell leads to J3
cell_pads	Cell leads on J3	—	2 · Solder the cell leads to J3
lra_wires	LRA leads	—	4 · Solder the LRA leads to J4
lra_pads	LRA leads on J4	—	4 · Solder the LRA leads to J4